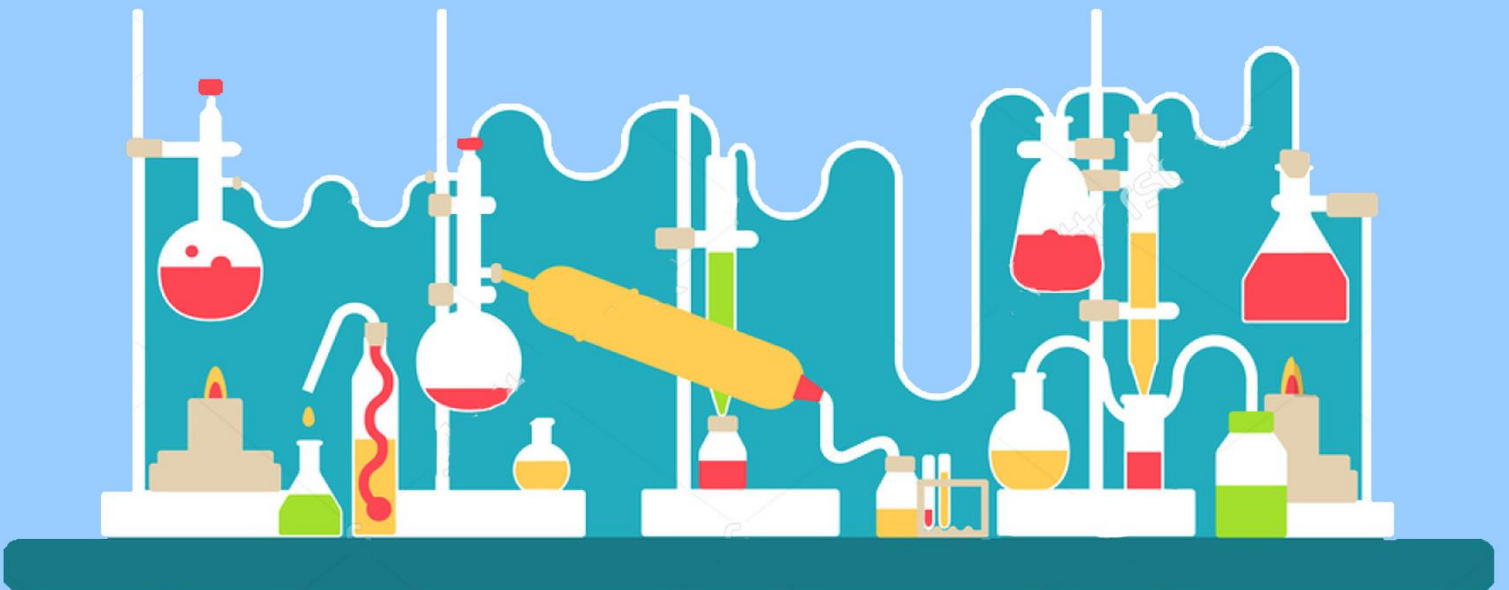




BIOLOGY LABORATORY



#7 Mitosis & Meiosis



General Notes:

- Most of cells are somatic cells have the ability to divide to produce identical daughter cells
- Germ line cells (sex cells) divide to produce gametes
- Cell division plays important roles in life:
 1. In prokaryotic: reproduction
 2. In eukaryotic: growth , development , regeneration , cell replacement , asexual reproduction

Cell cycle

Definition: **The cell cycle** or **cell-division cycle** is the series of events that take place in a cell leading to its division and duplication of its DNA (DNA replication) to produce two daughter cells.

Cell Cycle phases : 1- Mitotic phase 2- Interphase

Important notes:

- Mitotic phase (M) which include both mitosis and cytokinesis is the shortest phase
- Interphase account for more than 90% of the cycle
- Interphase divide into sub-phases : G₁(first gap), S (synthesis), G₂ (second gap)
- During interphase the cell prepare itself for division

Interphases

G(0): A resting phase where the cell has left the cycle and has stopped dividing this might be temporary or permanent (such as muscle and neuron) it's not part of interphase

the cell reached the end state of development will no longer divide

1. G₁(first gap) :

The first phase within interphase, extend from the end of the previous M phase until the beginning of DNA synthesis, is called G₁

In this phase, the cell increases its supply of proteins, increases the number of organelles (such as mitochondria, ribosomes), and grows in size

The duration of G₁ is highly variable

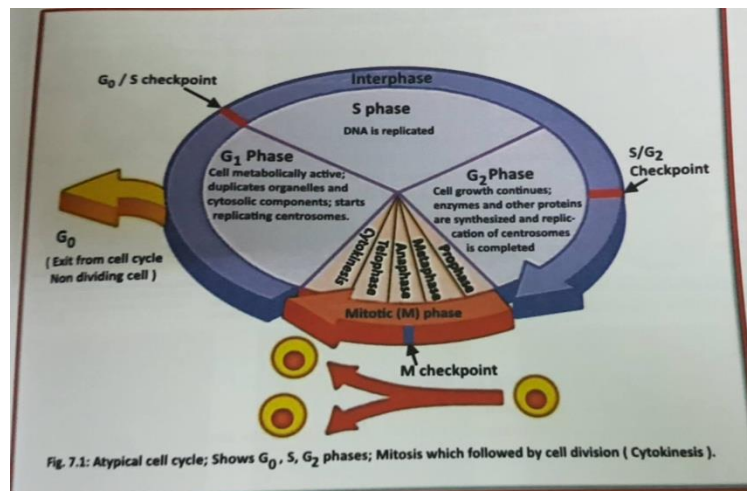
During this phase important cell cycle control mechanism activated during this period (G₁ checkpoint) to ensure everything is ready for DNA synthesis

2. S Phase (DNA replication): *during this phase, the amount of DNA in the cell has effectively doubled*

3. G_2 Phase (Growth)

is a period of protein synthesis and rapid cell growth to prepare the cell for mitosis.

There is another checkpoint (G2 checkpoint) to determine if the cell can proceed to enter M phase and divide or not.



MITOSIS

Mitosis is a part of the cell cycle when replicated chromosomes are separated into two new nuclei is often accompanied or followed by cytokinesis, which divides the cytoplasm, organelles and cell membrane into two new cells containing roughly equal shares of these cellular components

Stages of Mitosis :

- prophase,
- metaphase,
- anaphase,
- telophase

1- Prophase:

occurs after G₂ interphase

the cell prepares to divide by tightly condensing its chromosomes and initiating mitotic spindle formation, this process is called chromosome condensation

the onset of prophase, chromatin fibers condense into discrete chromosomes that are typically visible at high magnification through a light microscope. In this stage, chromosomes are long, thin and thread-like. Each chromosome has two chromatids. The two chromatids are joined at a place called centromere.

The nucleolus also disappears during early prophase.

Close to the nucleus of animal cells are structures called centrosomes, consisting of a pair of centrioles surrounded by a loose collection of proteins. The centrosome is the coordinating center for the cell's microtubules. A cell inherits a single centrosome at cell division, which is duplicated by the cell before a new round of mitosis begins, giving a pair of centrosomes. The two centrosomes polymerize tubulin to help form a microtubule spindle apparatus. The two centrosomes push these microtubules to opposite sides of the cell. Centrosomes are absent in plants

the nuclear envelope to disintegrate into small membrane vesicles. As this happens, microtubules invade the nuclear space. This is called *open mitosis*, and it occurs in some multicellular organisms. Fungi and some protists, such as algae or trichomonads, undergo a variation called *closed mitosis* where the spindle forms inside the nucleus, or the microtubules penetrate the intact nuclear envelope.

2- Metaphase:

Kinetochores (the place on either side of the centromere to which the spindle fibers are attached during cell division.)

- kinetochores microtubule begin search for and attach to chromosomal kinetochores
- spindle fiber are completely formed
- the chromosome become short and thick with two distinct chromatids
- the chromosomes to align along the *metaphase plate* or *equatorial plane*, an imaginary line that is centrally located between the two centrosomes (at approximately the midline of the cell)

3- Anaphase :

shortening of the kinetochore microtubules pulls the newly formed daughter chromosomes to opposite ends of the cell. During *anaphase B*, polar microtubules push against each other, causing the cell to elongate. chromosomes also reach their overall maximal condensation level.

4- Telophase:

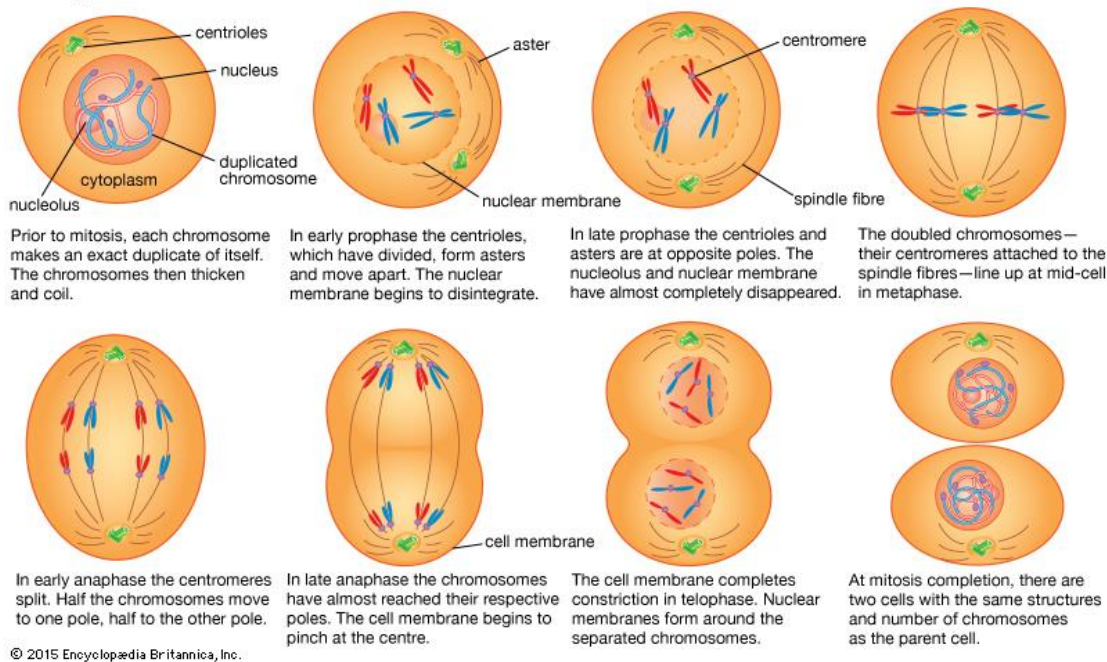
- reversal of prophase
- the daughter chromosome reached the opposite pole
- the chromosome undergo desperalization to form long, thread like structure called chromatine
- nucleolus appear
- new nuclear envelope forms using the membrane vesicles of the parent cell's old nuclear envelope
- the spindle fiber diappear

Cytokinesis

Cytokinesis is not a phase of mitosis but rather a separate process, necessary for completing cell division. In animal cells, a cleavage furrow (pinch) containing a contractile ring develops where the metaphase plate used to be, pinching off the separated nuclei. In both animal and plant cells, cell division is also driven by vesicles derived from the Golgi apparatus, which move along microtubules to the middle of the cell. In plants, this structure coalesces into a cell plate

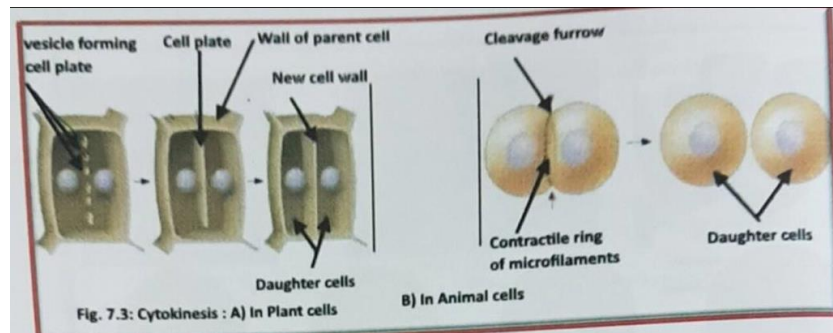
The direction from inside to outside

Mitosis, or somatic cell division



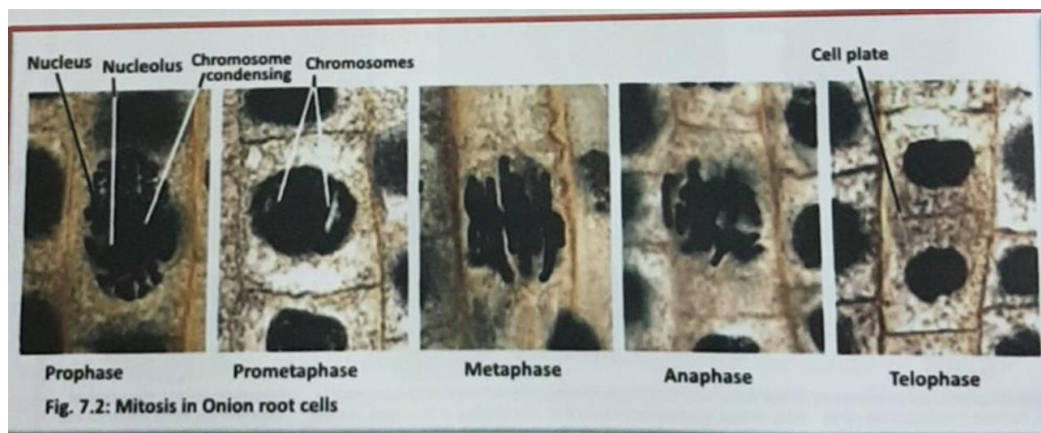
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Plant cell



(I) The cell cycle in plant cell

Obtain a prepared slide of a longitudinal section of an Allium (onion) root tip. Examine the square cells just inside the root cap. This is the root meristem (embryonic tissue) where mitosis is occurring. Farther up the study, root is the elongation zone, where cells are long rectangles; these cells are not undergoing mitosis. Also models available for each stage of mitosis and compare to the different mitotic stages represented in this figure

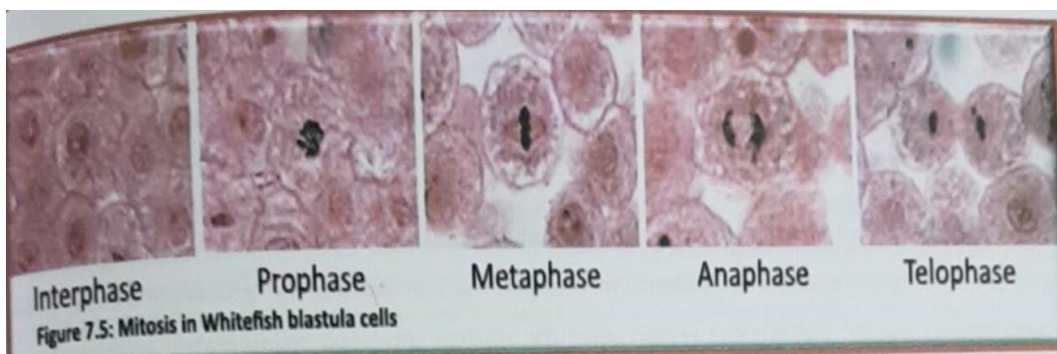


Animal cell

(II) the cell cycle in animal cells (Embryonic blastula stage of Whitefish)

The developing embryo of any organism is a good tissue to examine for mitosis, since cells must divide at a high rate to transform a fertilized egg (single cell) into the trillions of cells of a viable organism.

With several exceptions, mitosis in animals is like that in plants. One exception is that animal cells, unlike plant cells, contain centrioles, which are barrel-shaped structures that work as microtubules organizing centers.



Meiosis

Meiosis is a specialized type of cell division that reduces the chromosome number by half.

This process occurs in all sexually reproducing single-celled and multicellular eukaryotes, including animals, plants, and fungi. In meiosis, DNA replication is followed by two rounds of cell division to produce four each with half the number of chromosomes as the original parent cell. The two potential daughter cells meiotic divisions are known as Meiosis I and Meiosis II. Meiosis is important for the transmission of traits from one generation to the next and for the production of gametes; the sperms and the oocytes, that's why meiosis is important for sexual reproduction. In the body cells of most eukaryotes, chromosomes exist in pairs called homologous chromosomes. These are two chromosomes, which are physically similar and contain genetic information for the same traits. Cells that have two copies of each chromosome are called diploid ($2N$). (Gametes that are produced at the end of meiosis have only one copy of homologs and are called haploid ($1N$).

Meiosis consists of two divisions :

- A. During meiosis I homologous pairs of chromosomes are separated and cells become haploid
- B. During meiosis II the sister chromatids of each chromosome are separated (as in Mitosis).

Meiosis I

At the start of meiosis, each chromosome consists of two sister chromatids, connected at the centromere.

However, meiosis differs from mitosis, in that homologous pairs of chromosomes come together at the start of the meiosis, in a process called synapsis, forming a tetrad or bivalent.

Prophase 1

Just as in mitosis, the nucleolus and nuclear membrane disappear, the centriole which forms the spindle, get a part from each other and move to the opposite pole. As in mitosis, the chromosomes coiled and come thick, Visible, and each one contains two sister chromatids held together by centromere.

The chromosomes approximately line up with each other into homologous chromosome pairs. At this stage the synapsis (pairing/coming together) of homologous chromosomes takes place, facilitated by assembling of central element of the **synaptonemal complex**. Individuals of a pair are equal in length and in position of the centromere. Thus pairing is highly Specific and exact. The paired Chromosomes called **bivalent or tetrad chromosome**.

Crossing over occurs, in which non-sister chromatids of homologous chromosomes may exchange segment over regions of homology, and at the sites where exchange happens. Chiasmata

chiasma (pl. **chiasmata**) is the point where two homologous non-sister chromatids exchange genetic material during chromosomal crossover in meiosis (sister chromatids also form chiasmata between each other (also known as a chi structure)

The **synaptonemal complex** degrades and homologous chromosomes separate from one another a little, but the homologous chromosomes of each bivalent remain tightly bound at chiasmata.

at late of prophase I, chromosomes condense further and the four parts of the tetrads are actually visible.

The microtubules invade the nuclear region after the nuclear envelope disintegrates, attaching to chromosomes at the kinetochore. There are four kinetochores on each tetrad, but the pair of kinetochore on each sister chromatid fuses and functions as a unit during meiosis I.

metaphase I:

the paired homologous chromosomes align along an equatorial plane of the cell. There are random orientation of each bivalent along the metaphase plate, with respect to the orientation of the other bivalents along the same equatorial line. This is called **independent assortment**. It's important for genetic variability.

Definition of INDEPENDENT ASSORTMENT

formation of random combinations of chromosomes in meiosis and of genes on different pairs of homologous chromosomes by the passage according to the laws of probability of one of each diploid pair of homologous chromosomes into each gamete independently of each other pair

Anaphase 1: The homologous pairs of chromosomes separate from each other, as the Kinetochore microtubule shorten, it pulls one of each pair to the opposite ends of the cell.

Non-kinetochore microtubules lengthen, pushing the centrosomes farther apart. The cell elongates in preparation for division down the center.

Telophase 1: The first meiotic division effectively ends when the chromosomes arrive at the poles. Nuclear envelope and nucleolus reappear.

Cytokinesis, the pinching of the cell membrane in animal cells or the formation of the cell wall in plant cells occurs, completing the creation of two daughter cells. Each daughter cell now has half the number chromosomes but each chromosome consists of a pair of chromatids.

Meiosis II

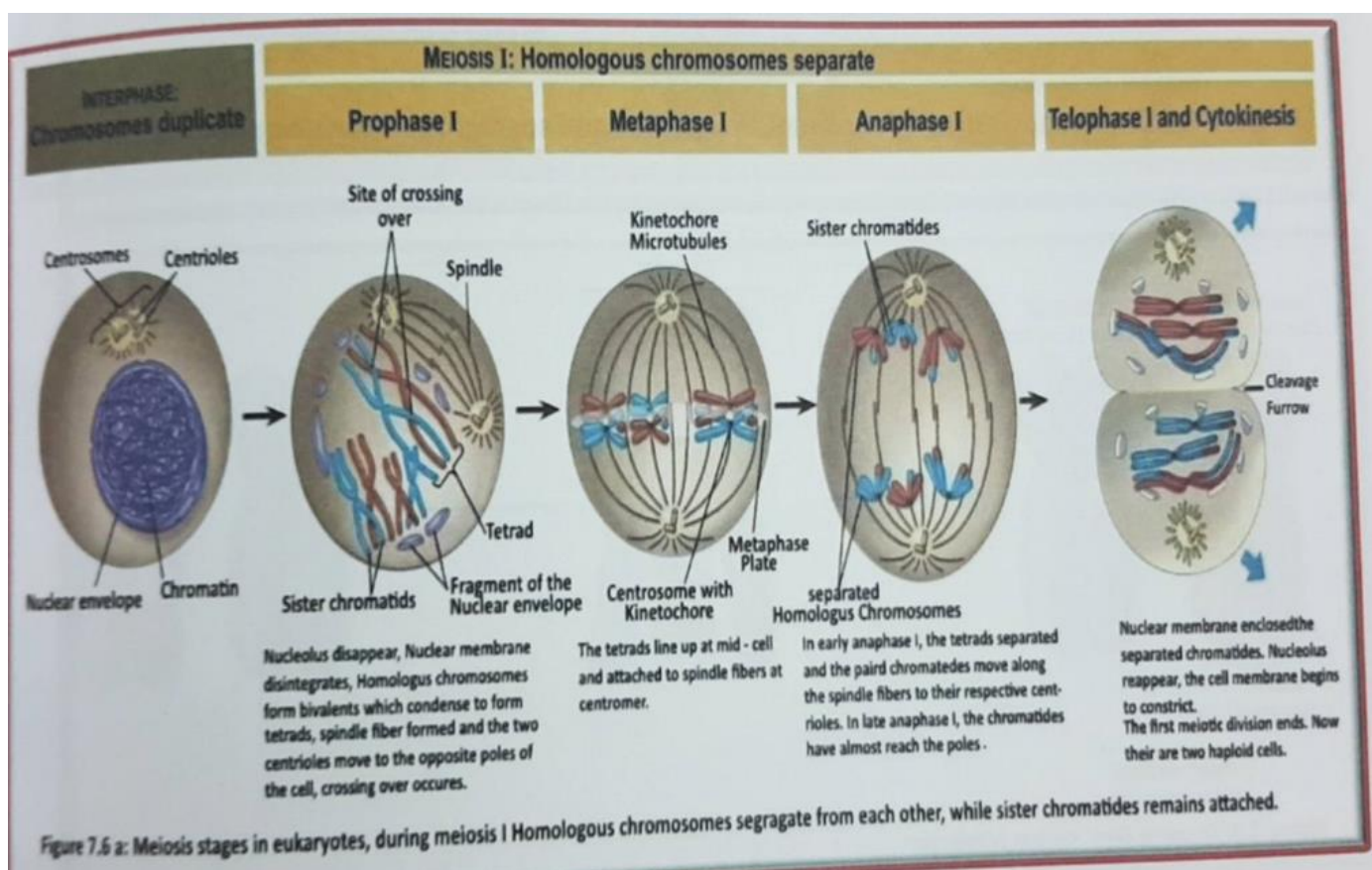
Meiosis II is the second meiotic division, and usually involves equational segregation, or separation of sister chromatids. Mechanically, the process is similar to mitosis, though its genetic results are fundamentally different. The end result is production of four haploid cells (n chromosomes, 23 in humans) from the two haploid cells (with n chromosomes, each consisting of two sister chromatids) produced in meiosis I. The four main steps of Meiosis II are: Prophase II, Metaphase II, Anaphase II, and Telophase II.

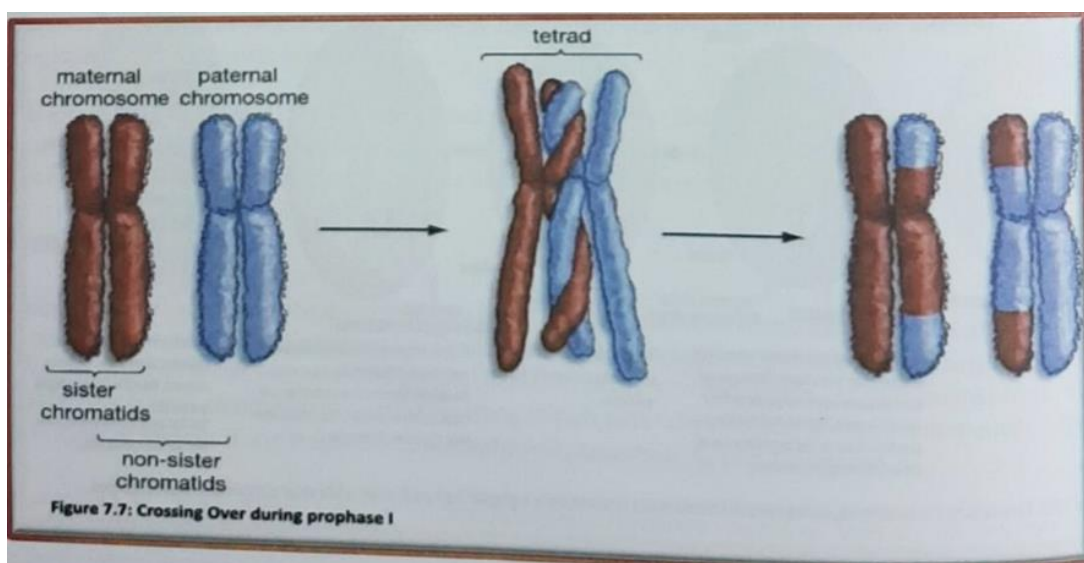
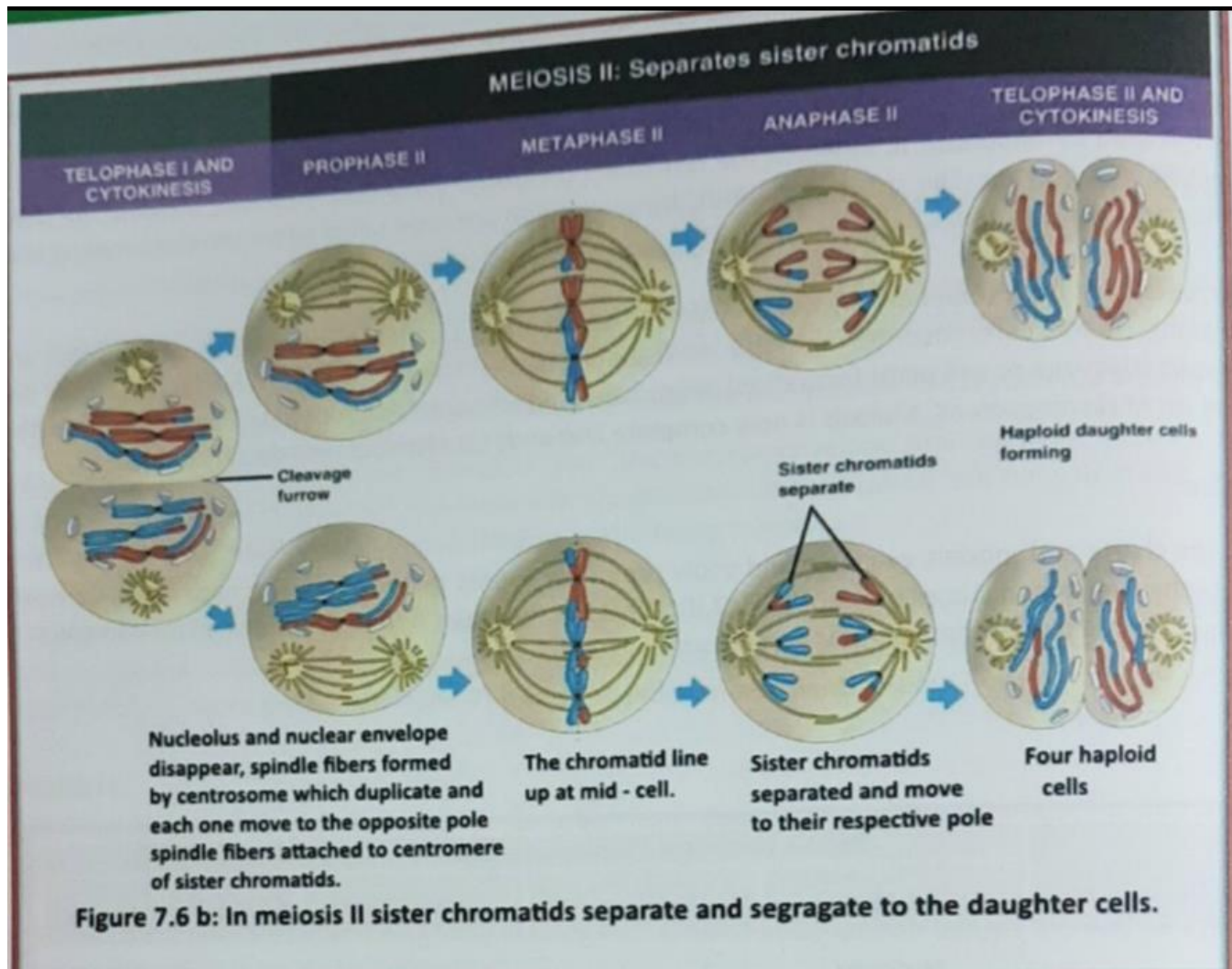
In prophase II: the nucleoli and the nuclear envelope disappear again as well as the shortening and thickening of the chromatids. Centrosomes move to the polar regions and arrange spindle fibers for the second meiotic division.

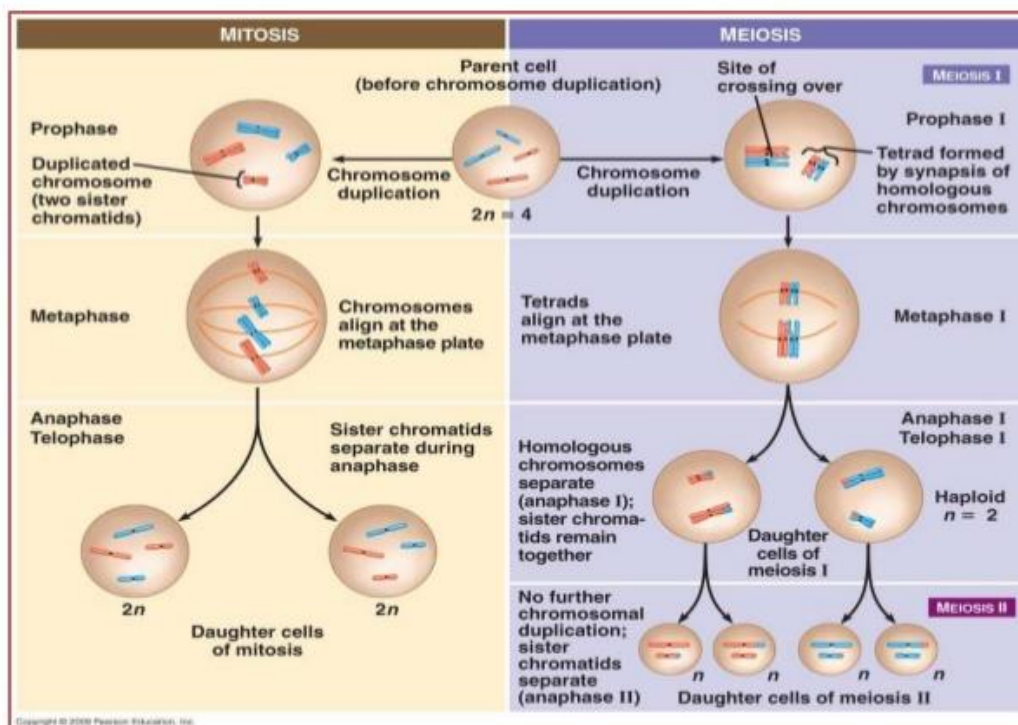
centrosomes in metaphase 2 the centromeres contain two kinetochores that attach to spindle fiber from the at opposite poles. The new equatorial metaphase plate is rotated by 90 degrees when, compared to meiosis I, perpendicular to the previous plate

This is followed by anaphase II, in which the remaining centromeric cohesion is cleaved allowing the sister chromatids to segregate. The sister chromatids by convention are now called sister chromosomes as they move toward opposing poles

The process ends with telophase II, which is similar to telophase I, and is marked by De-condensation and lengthening of the chromosomes and the disassembly of the spindle. Nuclear envelopes reform and cytokinesis (cleavage or cell plate formation) eventually produces a total of four daughter cells, each With a haploid set of chromosomes. Meiosis is now complete and ends up with four new daughter cells.







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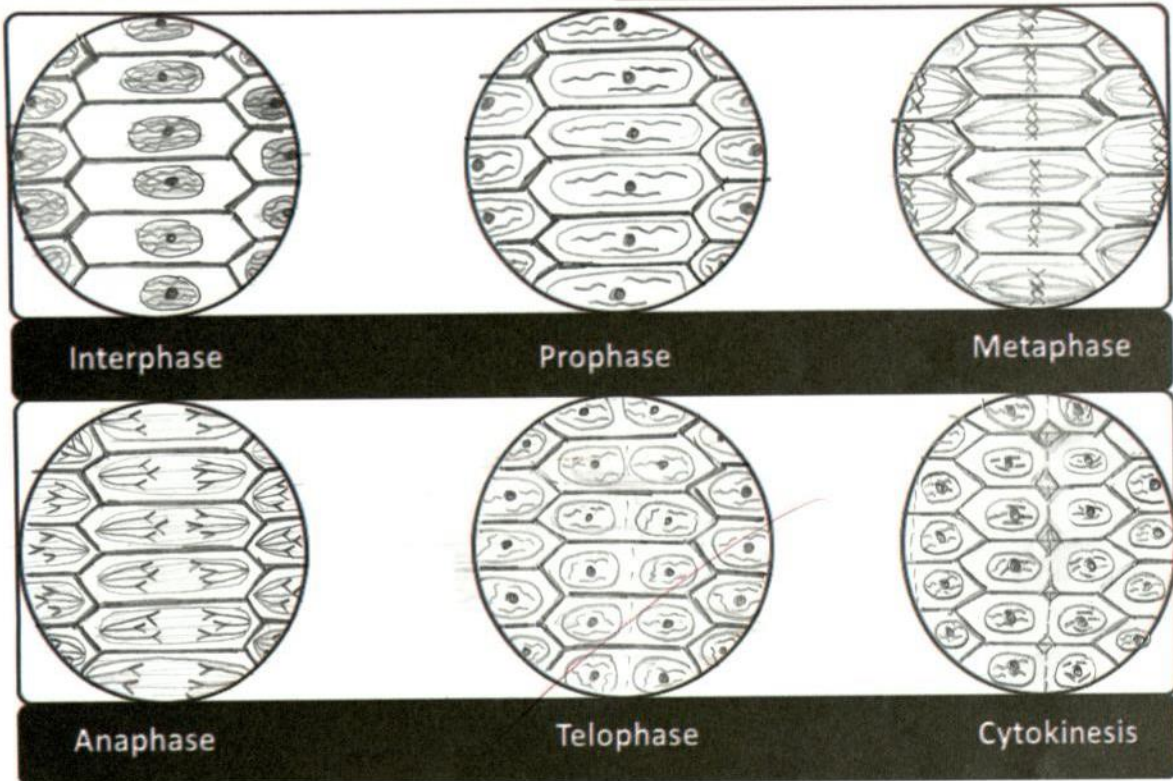
Difference between mitosis and meiosis	
Mitosis	Meiosis
This type of division takes place in somatic cells.	This type of division takes place in gametic cells.
Two daughter cells are formed.	Four daughter cells are formed.
No. of chromosomes remains diploid in daughter cells.	Number of chromosomes becomes haploid in daughter cells.
Mitosis is necessary for growth and repair.	Meiosis is necessary for sexual reproduction.
Crossing over does not take place.	Crossing over takes place.

اللاب فيه الكثير من التفاصيل ، مش غلط انك تعيد عليه مرة ثانية بعد يوم او يومين راح يساعدك على تثبيت دراستك >3 في ملف اللاب القادم سوف نضع لكم اختبار للابن معاً

report solution :

LABORATORY REPORT Mitosis and Meiosis

* Study the Onion Root tip then draw the following and label each part: Draw as you see under the Microscope only



- What are the function of Mitosis and Meiosis?

Mitosis	Meiosis
Development from fertilized egg	Produce gametes (animals) and oocytes.
Growth	Produce spores (sporophyte plant)
Repair (regeneration)	reduce number of chromosomes set by half
reproduction of unicellular organisms/prokaryotic	introduce genetic variability among gametes or spores

- List six differences between Mitosis and Meiosis:

Page 2 of mitosis Report

Mitosis	Meiosis
1. <u>one division</u>	1. <u>Two divisions</u>
2. <u>no synapsis</u>	2. <u>synapsis occur during prophase I.</u>
3. <u>Two daughter cells</u>	3. <u>Four daughter cells</u>
4. <u>produce diploid cells</u>	4. <u>produce haploid cells</u>
5. <u>no crossing over</u>	5. <u>crossing over during</u> prophase I
6. <u>identical to parent cell</u> daughter cells → identical to each other	6. <u>not identical to parent cell</u> not identical to each other

- Compare between Spermatogenesis and Oogenesis:

According to	Spermatogenesis	Oogenesis
Cell division	meiosis	meiosis
Human sex	male	female
Primary cell	spermatocyte	Oocyte
No. of chromosomes in Primary cells	46	46
End products	4 sperms	1 egg 3 polar body
No. of chromosomes in each cell produced	23	23
Genetic Variation	Yes, crossing over	Yes, crossing over